Assignment: Fundamentals of Linear Algebra for Artificial Intelligence

Instructions

- Answer all the questions.
- Show all intermediate calculations wherever applicable.
- Use a calculator if required.
- Round numerical answers to **two decimal places** where necessary.
- Clearly write the final answer for each question.

Part A – Scalars, Vectors and Matrices

Question 1

Identify whether each of the following is a **Scalar**, **Vector**, **Matrix**, or **Tensor**.

a) Scalar

$$5$$

b) Vector

$$\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

c) Matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

d) Tensor.

A colour image of size

$$256 \times 256 \times 3$$

Question 2

Determine the dimensions of the following matrices.

a)

Dimension = Rows x Columns.

2x3. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

b)

4x1 $\begin{bmatrix} 5 \\ 8 \\ 9 \\ 2 \end{bmatrix}$

c)

3x2. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$

Question 3

Given

$$A = \begin{bmatrix} 2 & 5 \\ 3 & 7 \end{bmatrix} \quad \begin{bmatrix} 4 & 1 \\ 6 & 8 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 4 & 1 \\ 6 & 8 \end{bmatrix}$$

Find

- $A + B$
- $A - B$
- $2A$

$A+B = \begin{bmatrix} 6 & 6 \\ 9 & 15 \end{bmatrix}$ $A-B = \begin{bmatrix} -2 & 4 \\ -3 & -1 \end{bmatrix}$ $2A = \begin{bmatrix} 4 & 10 \\ 6 & 14 \end{bmatrix}$.

Question 4

Find the transpose of the matrix

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$$

$\begin{bmatrix} 2 & 8 \\ 4 & 10 \\ 6 & 12 \end{bmatrix}$

Question 5

Find the transpose of

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 4 & 6 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 4 \\ 3 & 5 & 6 \end{bmatrix}$$

Part B – Matrix Multiplication

Question 6

Multiply

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} (1\times5)+(2\times7) & (1\times6)+(2\times8) \\ (3\times5)+(4\times7) & (3\times6)+(4\times8) \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

Question 7

Determine whether the following matrix multiplications are possible. If they are possible, write the dimensions of the resulting matrix.

a)

Possible.

2×4.

$(2 \times 3)(3 \times 4)$ $n = P$.

$(m \times n)(P \times q)$

b)

Not Possible

$(3 \times 2)(4 \times 3)$

c)

Possible

(4×5)

$(4 \times 2)(2 \times 5)$

d)

Possible.

(5×2)

$(5 \times 4)(4 \times 2)$

Question 8

Given

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

and

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \qquad = \begin{bmatrix} (2\times5)+(1\times7) & (2\times6)+(1\times8) \\ (4\times5)+(3\times7) & (4\times6)+(3\times8) \end{bmatrix}$$

Find

$$AB \qquad = \begin{bmatrix} 17 & 20 \\ 41 & 48 \end{bmatrix}$$

Question 9

Verify whether matrix multiplication is commutative.

Compute

$$AB \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{pmatrix} (1\times5)+(2\times7) & (6+16) \\ 15+28 & 18+32 \end{pmatrix}$$

and

$$BA \qquad = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \qquad AB \neq BA.$$

Based on your calculations, what do you conclude? Matrix Multiplication is not commutative.

Part C – Vector Operations

Question 10

Find the magnitude of the vector

Magnitude is length of vector.

$$x = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \qquad = |1\times1| = \sqrt{3^2+4^2}$$

$$= \sqrt{25} = 5.$$

Question 11

Find the magnitude of

$$y = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$$

$$\|y\| = \sqrt{2^2 + 5^2 + 6^2}$$

$$= \sqrt{4 + 25 + 36} = \sqrt{65} = 8.06.$$

Question 12

Compute the dot product of

$$x = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$$

$$x \cdot y = 10 + 8 + 3$$

$$= 21$$

and

$$y = \begin{bmatrix} 5 \\ 2 \\ 3 \end{bmatrix}$$

Question 13

Determine whether the following vectors are orthogonal.

dotproduct = 0 is orthogonal.

$$u = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$u \cdot v = 2 - 2$$

$$= 0.$$

Question 14

Normalize the vector

$$x = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$|x| = \sqrt{6^2 + 8^2} = \sqrt{100} = 10.$$

$$\begin{bmatrix} 6/10 \\ 8/10 \end{bmatrix} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}.$$

Question 15

Find the Euclidean distance between

$$A = (2, 3)$$

and

$$B = (8, 11)$$

$$\sqrt{(8-2)^2 + (11-3)^2}$$

$$\sqrt{6^2 + 8^2} = \sqrt{36 + 64}$$

$$\sqrt{100} = 10.$$

Part D – Vector Spaces and Span

Question 16

Determine whether the following vectors are linearly independent.

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad = 2\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Explain your answer. They are linearly dependent, because V_2 vector is the scalar multiple of V_1.

Question 17

Determine whether v belongs to span of V_1 & V_2.

$$v = \begin{bmatrix} 5 \\ 7 \end{bmatrix} \qquad a\begin{bmatrix} 1 \\ 1 \end{bmatrix} + b\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

belongs to the span of

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad a + 2b = 5$$

$$a + b = 7$$

and

$$b = -2, \quad a = 9.$$

$$v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad 9\begin{bmatrix} 1 \\ 1 \end{bmatrix} - 2\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

Question 18

Find a basis for $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ Rank 2 → 2 linearly independent vectors.

$\mathbb{R}^2$

Question 19

Find the rank of the matrix 1 (as the second row is linearly dependent.

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Question 20

Find the rank of 2

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Part E – Orthogonality

Question 21

State whether the following pairs of vectors are orthogonal.

a) Yes, dot product is zero.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

b) Yes, dot product is zero.

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

c) No, dot product is non zero.

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

Question 22

Find the cosine similarity between

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

and

$$y = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Dot product = 11

Magnitude - $\sqrt{1^2+2^2} = \sqrt{5}$

$\sqrt{9+16} = \sqrt{25} = 5.$

$\cos\theta = \frac{11}{5\sqrt{5}} \simeq 0.98.$

Interpret the result.

Question 23

Explain, in your own words, the difference between:

- Orthogonal vectors – Perpendicular, dot product is zero.
- Orthonormal vectors → Perpendicular, unit vectors.
- A basis – linearly independent vectors → building blocks of space.
- A vector space – full set of vectors.